Cafe: Improved Federated Data Imputation by Leveraging Missing Data Heterogeneity

Sitao Min, Hafiz Asif, Xinyue Wang, and Jaideep Vaidya, *Fellow, IEEE*

***Abstract*—Federated learning (FL), a decentralized machine learning approach, offers great performance while alleviating autonomy and confidentiality concerns. Despite FL's popularity, how to deal with missing values in a federated manner is not well understood. In this work, we initiate a study of federated imputation of missing values, particularly in complex scenarios, where missing data heterogeneity exists and the state-of-the-art (SOTA) approaches for federated imputation suffer from significant loss in imputation quality. We propose Cafe, a personalized FL approach for missing data imputation. Cafe is inspired from the observation that heterogeneity can induce differences in observable and missing data distribution across clients, and that these differences can be leveraged to improve the imputation quality. Cafe computes personalized weights that are automatically calibrated for the level of heterogeneity, which can remain unknown, to develop personalized imputation models for each client. An extensive empirical evaluation over a variety of settings demonstrates that Cafe matches the performance of SOTA baselines in homogeneous settings while significantly outperforming the baselines in heterogeneous settings.**

***Index Terms*—Federated learning, missing data imputation, data quality, data heterogeneity.**

I. Introduction

TODAY, data is collected by commercial, governmental, and even non-governmental organizations in all settings. Most often, this data is naturally collected at various institutional sites. For example, healthcare data may be collected by primary care physicians, hospitals, insurance companies, pharmacies, and other organizations. Due to a number of reasons, including regulatory compliance, institutional autonomy, confidentiality concerns, or even simply economic reasons, this data is not, and often cannot be, shared in raw form or processed centrally. Thus, we are unable to apply modern machine learning or artificial intelligence models directly over this data effectively prohibiting applications such as fraud detection, diagnosis of rare diseases, or identification of adverse effects [1], [2]. Federated Learning (FL), which allows the building of models over distributed data, is rapidly gaining popularity as an alternative to data centralization and the creation of data silos. In FL, models are locally built over the distributed data and then aggregated at a central site. This mitigates many of the concerns with maintaining institutional autonomy, raw data confidentiality, regulatory compliance, and even limits computational and communication costs, which are quite significant for large quantities of data (often required to train modern AI/ML models) [3], [4].

A critical preliminary step in machine learning over real data – whether centralized or federated – is data pre-processing or cleaning, in particular, handling the missing values in the data, which often occur in real life [5]. However, FL typically relies on complete data (i.e., without any missing values). As a result, each participating silo must first address its local missing data issues prior to engaging in the FL process. Traditionally, most existing FL studies have typically managed missing values through basic imputation techniques or by removing incomplete records locally and independently within each silo [6], [7]. Recent advancements have introduced a collaborative approach based on FL for imputation, known as federated imputation [8], [9], [10], [11], [12]. It enables silos to leverage information from the data of other participants during the imputation process. Studies have demonstrated that the federated imputation approach can improve the generalizability and robustness of imputed outcomes across silos [8], [9], [10], [11], [12].

However, existing studies of federated imputation predominantly focus on basic missing mechanisms (MCAR/MAR [13]) with limited focus on complex mechanisms such as Missing Not at Random (MNAR). This is despite the fact that real-world scenarios indicate that missing data mechanisms are often complex and predominantly MNAR [14], [15], [16]. Furthermore, existing studies overlook a crucial characteristic of missing data in federated networks – the heterogeneity of missing data mechanisms across participating silos.

In real-world scenarios, especially in the healthcare domain, missing data mechanisms can vary significantly across silos, which can lead to missing data heterogeneity. These variations can be due to individual and organizational factors. At the individual level, missing data can be due to privacy concerns [17], the level of trust and acceptability that individuals perceive at an institution [16], and their health-related and socioeconomic status, e.g., the severity of health-condition, race, and gender [18], [19], [20]. At the organizational level, missingness in healthcare data can arise from administrative policies, bad data collection protocols, or documentation practices within healthcare facilities,

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Sitao Min and Jaideep Vaidya are with Rutgers University, Newark, NJ 07102 USA (e-mail: sm2370@business.rutgers.edu; jsvaidya@rutgers.edu).

Xinyue Wang is with Center for Applied Statistics and School of Statistics, Renmin University of China, Beijing 100872, China (e-mail: xinyue.wang@ruc.edu.cn).

Hafiz Asif is with Rutgers University, Newark, NJ 07102 USA, and also with Hofstra University, Hempstead, NY 11549 USA (e-mail: hafiz.asif@hofstra.edu).

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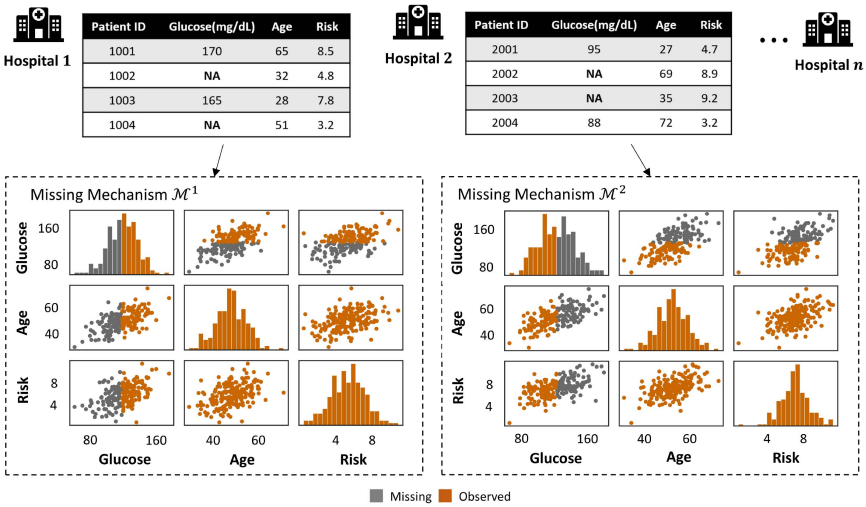


Fig. 1. Heterogeneous non-MCAR missing data Example. Left and right panels show data from two hospitals in a distributed healthcare network (DHN), with Glucose measurements missing due to heterogeneous mechanisms ($\mathcal{M}^1$, $\mathcal{M}^2$). Histograms compare observed (orange) and missing (gray) value distributions, while scatter plots highlight differences in correlations. Even though hospitals' data is iid, heterogeneous mechanisms render observed data (e.g. Glucose) non-iid. This example also illustrates that Hospital 1's observed data distribution of Glucose can potentially inform the imputation of Hospital 2's Glucose missing data and vice versa, this is referred to as complementarity. This type of complementarity motivates the potential of personalized FL framework to enhance imputation quality.

as well as from the clinical practices of physicians and medical staff [16], [21], [22]. For instance, Zachariasse et al. [23] found missing triage data in three European emergency departments were due to staff shortages and bad clinical protocols, and Wells et al. [21] reported missing clinical data at Cleveland Clinic were due to non-digitized paper records. Additional examples of different mechanisms for missing data in healthcare settings can be found in Zamanian et al. [16].

Consequently, assuming homogeneous missing mechanisms across silos in federated networks is neither practical nor reasonable.

Notably, heterogeneity is already recognized as a fundamental challenge in practical FL applications, covering aspects such as data, model, and system heterogeneity [4], [24], [25], [26]. Similar to the challenges posed by local data heterogeneity in traditional FL algorithms (e.g., Federated Averaging [27]) for prediction models, heterogeneity in missing data can also pose significant challenges for federated imputation. Existing solutions [8], [9], [11] that rely on simple aggregation of local imputation models into a single global model can suffer from a significant loss in imputation quality under complex heterogeneous missing data scenarios (as seen in Section IV). Thus, missing data heterogeneity remains an underexplored area in current research.

Therefore, in this article, we consider the *heterogeneous complex missing data setting, where* 1) n clients (i.e., organizations collecting data) have independent and identically distributed (iid) data samples ($\boldsymbol{X}^k$ for client k) possibly with missing values, and, 2) the missing values generating mechanism [28]) is complex, i.e., mechanism $\mathcal{M}^k$ for client k depends on data that is not observed (MNAR), and 3) missing mechanism are heterogeneous, i.e., $\mathcal{M}^k$ can differ across clients. Below, we provide an example as shown in Fig. 1 to allow the reader to understand the setting better.

Example: Multiple hospitals in a Distributed Health Network (DHN) aim to predict diabetes risk through federated learning. Some hospitals (e.g., Hospital 1) predominantly record only aberrant diagnostic (e.g., glucose levels) values from doctor's notes due to data documentation practices [21], leaving normal values for less severe patients missing from its records. Other hospitals (e.g., Hospital 2) have values of abnormal glucose levels missing, as patients with abnormal glucose levels are unwilling to disclose their past diagnostics due to low trust in care quality [29], [30], [31]. In this case, the underlying mechanisms generating missing values are heterogeneous and complex.

In this study, we propose a novel federated imputation method motivated by a key insight: the heterogeneity in missingness manifests as key characteristics in missing and observable data across silos, which, if recognized, can be leveraged to benefit the local imputation of clients' missing data. Specifically, our method models and learns an approximation of the missing mechanism for each client (for each feature) locally, and then uses their pairwise similarity (referred to as *pairwise complementarity*), as a measure of the degree of heterogeneity of the mechanisms. The clients' pairwise complementarity, along with some additional information, is then used to compute *complementarity adjusted federated (*Cafe*) averaging* of the local imputation models as a personalized federated imputation solution for heterogeneous complex miss- ing data. Also, while we do not focus on privacy, since (as will be discussed later), only model parameters of a missing mechanism prediction model are transferred from clients to the server, this significantly limits information disclosure during imputation. To summarize, this article makes the following key contributions.

- Initiates the study of how to leverage heterogeneity in missing data across silos to improve data imputation.
- Proposes the notion of a missing mechanism complementarity to measure the heterogeneity in the missing data of different datasets and develops an efficient method based on a missing mask prediction model integrated with Impu-

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tation via Chained Equations (ICE) algorithm [32], [33]) to compute it in the federated setting.
- Develops a new method of complementarity adjusted federated (*Cafe*) averaging, which is applied to learn imputation models. Importantly, this is the first imputation method that can naturally and gracefully handle the case of MNAR missingness. Using *Cafe*, we develop and implement *Cafe* averaging-based federated ICE imputation, referred to in the rest of the paper as *Cafe* ICE.
- Conducts an extensive evaluation of *Cafe* ICE over a range of medical datasets, data distribution settings, and non-MCAR missing mechanisms. The results show that our method outperforms local imputation, simple averaging based Federated ICE as well as other state-of-the-art imputation methods over various performance metrics.

The rest of this paper is structured as follows. Section II provides key notations, definitions, and preliminary background to contextualize our work. Section III presents the problem setting as well as our proposed approach. We present the results of a comprehensive empirical evaluation in Section IV. Section V presents the related work. Finally VI concludes the paper and discusses future work.

II. Preliminaries

We start by providing the key notations, definitions, and the background needed to place our work in context.

Data, Missingness, and Clients: $\mathcal{X}^* = \prod_{f\in[d]} \mathcal{X}_f^*$ denotes the data universe of d features possibly *with missing values*, i.e., the values of one or more features may be missing in a data sample from $\mathcal{X}^*$; here $\mathcal{X}_f^* = \mathcal{X}_f \cup \{\texttt{NA}\}$ where NA denotes that the value is missing and $\mathcal{X}_f$ is the set of possible values of feature f ($[d] = \{1, 2, \ldots, d\}$). We use $\mathcal{X} = \prod_{f\in[d]} \mathcal{X}_f$ to denote the data universe *without* missing values. Note that we only consider numeric features in this work and refer to them with labels $1, 2, \ldots, d$, though our work is extensible to categorical data.

We consider n parties, referred to as *clients*, where each client $k = 1, \ldots, n$ receives (or has) data $\boldsymbol{X}^k$ of N^k samples from $\mathcal{X}^*$, that is, one or more samples in $\boldsymbol{X}^k$ have NA as the value for one or more features. Through the work, when k is not quantified, it refers to an arbitrarily chosen client unless mentioned otherwise. Now, for the data $\boldsymbol{X}^k$, $\boldsymbol{X}^k_{fi}$ denotes the values of feature f for the i-th sample in $\boldsymbol{X}^k$, while $\boldsymbol{X}^k_f$ denotes the values of f for all the samples in $\boldsymbol{X}^k$. And, $\boldsymbol{X}^k_{-f}$ denotes the data for all the features but f. Note that for simplifying the exposition we use the convention that the *first subscript* for the data refers to the feature.

We use $\boldsymbol{R}^k$ to denote the *missing indicators* of the data $\boldsymbol{X}^k$, that is, $\boldsymbol{R}^k_{fi} = 1$ if $\boldsymbol{X}^k_{fi} = \texttt{NA}$ and $\boldsymbol{R}^k_{fi} = 0$ otherwise.

We say that a feature f is *incomplete* in $\boldsymbol{X}^k$ if one or more samples in $\boldsymbol{X}^k$ are missing values for f; similarly, f is *incomplete* in $(\boldsymbol{X}^1, \ldots, \boldsymbol{X}^n)$ if f is incomplete for at least one client.

A sample in $\boldsymbol{X}^k$ is *f-unobservable* if the the value of the feature f is missing (i.e., is NA) in the sample; otherwise the sample is called *f-observable*.

Missing Mechanisms: The clients data is from $\mathcal{X}^*$ although the original data comes from $\mathcal{X}$ (i.e., without any missing values). We use the standard approach of missing mechanism to describe this transformation, i.e., the data without-missing values to with-missing values [13]. A missing mechanism $\mathcal{M}_f$ for a feature f takes the data from $\mathcal{X}$ and outputs data in $\mathcal{X}^*$, that is, $\mathcal{M}_f$ (which can be probabilistic) generates missing values by replacing certain values of f by NA's.

Missing mechanisms are categorized into two types based on how the generated missing values are related to the data. These categories are as follows.
- *Missing Completely At Random (MCAR):* We say a missing mechanism is MCAR if the distribution of missing values in the data is *in*dependent of the data distribution.
- *Non-MCAR:* We say a missing mechanism is non-MCAR if the distribution of missing values depends on the data distribution. There are however two types of non-MCAR mechanisms: 1) when missing data distribution depends on the observable data distribution (i.e., the feature that the client observes or collects), we say the mechanism is missing at random (MAR); 2) when missing data distribution depends on the unobservable data distribution (i.e., the feature that the client does not observe or collect), we say the mechanism is missing not at random (MNAR).

In this work, we will focus on non-MCAR mechanisms, i.e., we assume that the missing values in feature f in client k's data are generated by a non-MCAR mechanism $\mathcal{M}^k_f$. We note that the non-MCAR settings — in particular, MNAR settings, also referred to as non-ignorable missingness — are challenging to tackle as opposed to MCAR settings [13].

Imputation via ICE in the Centralized Setting: In data analysis and machine learning, complete datasets are crucial. A common method to address missing values is imputation. The MICE/ICE (Imputation by Chained Equations) methods [32], [33], [34] are popular for their effectiveness, simplicity, and efficiency. ICE employs feature-specific imputation models that are built based on inter-feature correlations to impute missing values. It then leverages iterative refinement of missing value estimates to capture the joint distribution of missing data across multiple features, thereby ensuring accurate imputations under complex missing data scenarios. ICE typically uses linear regression to learn imputation models, which is both effective and efficient, making it a preferred choice for handling missing data, though it can be modified to use neural networks and random forests as well [35]. Here, we consider the following ICE approach (simplified from MICE) from [8] in the centralized setting, where all the data is available at one site. Fig. 2 illustrates the details of the ICE approach.

Initialization: The missing values are initially filled in with simple imputations, like the mean or median.

Iterative Imputation Process: Each iteration consists of imputing each incomplete feature f in a sequential manner.

a) *Learning imputation model for f:* The algorithm treats f as the target and the remaining features as predictors. A linear regression model is then fitted on the f-observable samples of the target feature.

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TABLE I
LIST OF KEY SYMBOLS AND THEIR DEFINITIONS

Category	Symbol	Definition
Setting	$\boldsymbol{X}^k$	local data of client k
	N^k	sample size of local data of client k
	$\boldsymbol{R}^k$	missing indicator of local data of client k
	$\mathcal{M}_f^k$	missing mechanism of client k
	$\widetilde{\boldsymbol{X}}^k$	imputed data of client k
	f unobservable sample	the sample whose value of the feature f is missing
	f observable sample	the sample whose value of the feature f is not missing
	$\boldsymbol{A}_f^k$	target for fitting imputation model of feature f for client k
	$\boldsymbol{A}_{-f}^k$	predictors for fitting imputation model of feature f for client k
	$\boldsymbol{B}_f^k$	imputed value for missing data using imputation model of feature f and client k
	$\boldsymbol{B}_{-f}^k$	predictors for imputation model of feature f and client k used to generate imputed values
Algorithm	μ_f^k	local mean for initial imputation of feature f for client k
	$\boldsymbol{\theta}_f^k$	parameters of imputation model of feature f for client k
	$\boldsymbol{\xi}_f^k$	parameters of mechanism prediction model of feature f for client k
	$w^{k\ell}$	aggregation weights for client k and ℓ
	$c^{k\ell}$	complementarity score between client k and ℓ
	s^ℓ	sample size score of client ℓ
	$\widetilde{\boldsymbol{\theta}}_f^k$	updated parameters of imputation model of feature f for client k
	α	hyperparameter controls the importance of sample size to complementarity
	γ	hyperparameter controls importance of client's model w.r.t others model
	β	hyperparameter controls normalization of aggregation weights

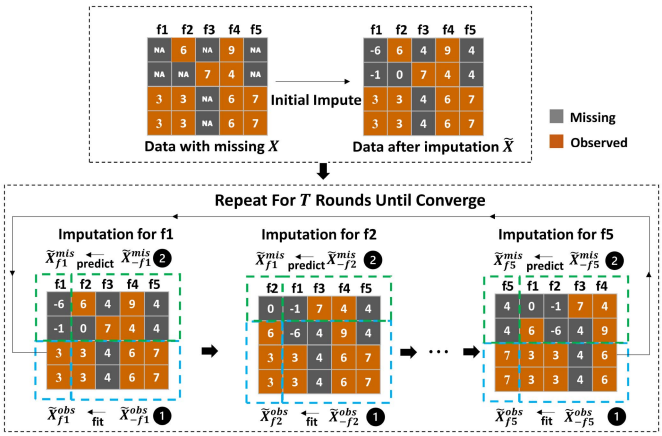


Fig. 2. ICE Algorithm, dataset with 5 features: f1 to f5.

b) *Updating imputations for f:* The missing values in f are then imputed (or updated) using the newly fitted model.
c) *Cycling through incomplete features:* (a) and (b) are repeated for each incomplete feature using the most recently imputed data for fitting the model in part (a).

Convergence: Once the process converges, meaning that subsequent imputations do not significantly change, the last dataset, with all missing values imputed, is the final imputed dataset.

Federated ICE: Federated learning (FL) enables building models over distributed data [36]. To create the federated version of ICE, the parts (a) and (b) of the iterative process (in the centralized ICE, described above) are done differently. Here, for each feature f, each client learns the imputation models on its data, and sends its model to a designated party, called the *aggregator*. The aggregator computes a global model by taking a weighted average of all the received models, and sends the global model back to the clients. Each of the clients uses the received global model and uses it to impute the feature f to prepare for the next incomplete feature or the round.

III. PROBLEM SETTING AND PROPOSED APPROACH

We now describe our problem setup as well as how different data-owning clients can use federated learning and leverage heterogeneity of their missing mechanisms to build better imputation models to boost the quality of their missing data imputation. We first outline the high-level approach, namely complementarity adjusted federated (*Cafe*) averaging, discuss its appropriateness, and then discuss the details. Table I shows all important symbols and their definitions related to our study.

A. Problem Setting

We consider n clients, each of which has iid samples from $\mathcal{X}^*$, the same data universe with missing values. Thus, the data, $\boldsymbol{X}^k$, of any client $k = 1, \ldots, n$, may have one or more features missing in multiple samples, i.e., incomplete features. We particularly consider the setting, where the mechanisms for at least one incomplete feature, say f, are non-MCAR and heterogeneous across clients. Namely, for instance, consider the mechanisms $\mathcal{M}_f^k$ and $\mathcal{M}_f^\ell$ (for $k \neq \ell$) that result in different (i.e., non-iid) distributions of f-observable samples. Furthermore, the differences in these distributions are helpful for imputation, i.e., observable data at one client provides useful information to model the missing data in that of the other — recall that we refer to this as complementary. Of course, in practice we expect to have varying level of complementarity, from none,

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to weak to strong. Thus, in our experimental evaluation, we will consider this range of heterogeneity. We aim to develop a federated learning imputation method that can leverage the complementarity in missing mechanism to improve data imputation both locally and globally. We note here that it is sufficient to use the distribution of the data, rather than the raw data itself, to estimate complementarity.

B. Vanilla Federated ICE and its Limitations

We now describe a basic federated imputation method and discuss the limitations of the existing approaches in dealing with heterogeneous non-MCAR missing mechanisms. Let $\mathcal{F}$ be the *set of incomplete features* across $\boldsymbol{X}^1, \ldots, \boldsymbol{X}^n$. To leverage the data federation, the clients collaborate in building the ICE imputation model over the federated data as follows. In each round $t = 1, \ldots, T$:

1) *Each client k updates its imputation model for each $f \in \mathcal{F}$:* Client k starts with the imputation models from the last round (one for each $f \in \mathcal{F}$), and learns the imputation models ($\boldsymbol{\theta}_f^k$ for every $f \in \mathcal{F}$) for the current round using its local data ($\boldsymbol{X}^k$). The client k then sends the updated imputation models to the aggregator.
2) *The aggregator performs federated averaging:* The aggregator receives the imputation models ($\boldsymbol{\theta}_f^1, \ldots, \boldsymbol{\theta}_f^n$ for each $f \in \mathcal{F}$) from all the clients. It then aggregates the models, e.g., by taking a uniform or *weighted average*, and sends the aggregated model back to the clients.

Note that for each feature all clients share the same global model in this approach. Federated averaging, is the most crucial step. Many innovative solutions have been constructed by considering different meaningful weighting schemes [36], [37], [38], [39]. In vanilla federated averaging, all the local models are weighted proportionally to their training samples (*TS*)-size [27]. However, simply using samples sizes is insufficient to completely capture the underlying mechanism heterogeneity and cannot result in the optimal model being selected.

We develop a novel weighting scheme, Cafe averaging, in which the weights are not only proportional to *TS*-sizes, but in addition, the scheme accounts for the heterogeneity of the missing mechanism and leverages it to adjust weights. This helps to mitigate the issues of non-iid training sample space, which is induced by the heterogeneity. We discuss this next.

C. Complementarity Adjusted Federated (Cafe) Averaging

Despite the fact that the data at each client is iid, the set of f-observable samples and f-*un*observable samples across the clients can be non-iid when the missing mechanisms are complex and heterogeneous and across clients. For instance, consider that the clients' missing mechanisms are complex (i.e., non-MCAR) and heterogeneous; this results in f-observable samples across clients to vary in their distribution and numbers (recall that in f-observable samples, the feature f is observable, i.e., *not* missing). A specific example of this can be observed in Fig. 1, where the observable data at Clients 1 and 2 are clearly non-iid due to the missingness mechanism.

The standard federated imputation setup for a feature f requires training over clients' data, which consist of f-observable samples, while the data to be imputed consist of f-*un*observable samples. Thus, in heterogeneous scenarios, f-observable and f-*un*observable samples can be non-iid not only at one client but also across clients. In such a case, a single global model, learned via federated averaging, has been shown to perform extremely poorly for local prediction [40], [41], which in our case is the imputation of a client's local f-unobservable samples. Personalization techniques are becoming popular to handle such heterogeneous scenarios, where instead of one global model, multiple models are learned, each personalized for a specific client. However, due to the unique nature of the imputation problem, the existing personalization techniques are not directly applicable for imputation, and hence, developing personalized federated imputation approaches is important.

Our personalization technique is inspired by the idea of complementarity; namely, that the differences in distributions of f-observable samples across clients can be helpful for imputation, i.e., f-observable data at one or more clients provides useful information to model the missing data in f-*un*observable samples of another client. Estimating this type of complementarity, however, is challenging in federated settings since data at one client is not directly available to another.

Thus, we propose a novel (missing mechanism) model-based approach to estimate complementarity, which we will use to build a personalized imputation model optimized for each client's local missing data distribution. In particular, we learn a model, $\boldsymbol{\xi}_f^k$, to approximate the missing mechanism ($\mathcal{M}_f^k$) for each incomplete feature f at each client k (We will discuss mechanism estimation model in detail in Section III-D). Then, for any pair of clients k and ℓ, we measure the *dissimilarity* between the two mechanism estimation models, i.e., $\boldsymbol{\xi}_f^k$ and $\boldsymbol{\xi}_f^\ell$, to assess how complementary are the f-observable and f-*un*observable data distributions across the two clients.

In addition to complementarity, the size of training data can also influence imputation model fitting [8], [42]. To account for this, we incorporate both complementarity and training sample size considerations into our personalized model aggregation through a weighted integration approach. Specifically, to adjust the weights for complementarity and training sample size (for an incomplete f), we need to compute weights, $w^{k\ell}$ for each pair of client k and ℓ ($k \neq \ell$), which we do as follows: for each client k, its own model ($\boldsymbol{\theta}_f^k$) gets the weight $\gamma \in [0, 1]$, while every other client's model, $\boldsymbol{\theta}_f^\ell$, gets the weight $(1-\gamma)w^{k\ell}$; here $w^{k\ell}$ is the complementarity and sample-size adjusted weight, and it is proportional to a weighted average of complementarity scores of the missing mechanism models ($\boldsymbol{\xi}_f^k$ and $\boldsymbol{\xi}_f^\ell$) and the number of f-observable samples. Namely, for $\alpha \in [0, 1]$,

$$w^{k\ell} \propto \alpha \begin{pmatrix} \text{No. of f-observable} \\ \text{samples at } \ell \end{pmatrix} + (1-\alpha) \begin{pmatrix} \text{complementarity b/w} \\ \boldsymbol{\xi}_f^k \text{ and } \boldsymbol{\xi}_f^\ell \end{pmatrix}$$

Thus, cafe average for client k is given as:

$$\texttt{Cafe-Avg}(k, f) = \gamma \cdot \boldsymbol{\theta}_f^k + (1-\gamma) \sum_{\ell \neq k} w^{k\ell} \cdot \boldsymbol{\theta}_f^\ell$$

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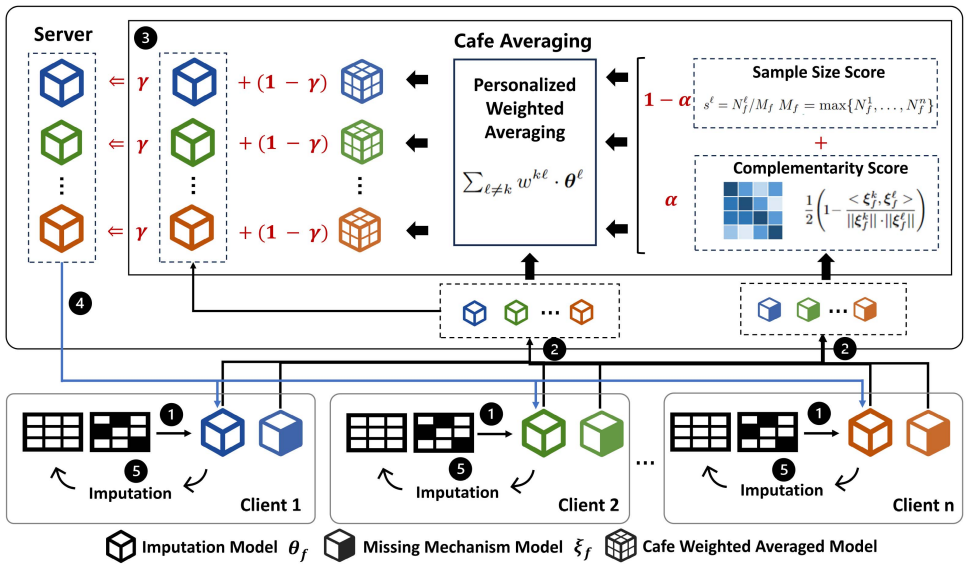


Fig. 3. Complementarity Adjusted Federated (*Cafe*) Imputation. This figure illustrates Cafe Averaging of imputation models from clients for one feature at each iteration. The whole process follows the following steps: (1) Clients fit the imputation and mechanism models, (2) Clients upload both models and mechanism to the server, and (3) Server conducts *Cafe* averaging. (4) Server dispatches personalized aggregated imputation models to the clients (5) Clients use updated imputation models to impute their local missing data, and continue with the next iteration, if necessary.

D. Complementarity Adjusted Federated (Cafe) ICE

Let us now discuss how we use Cafe averaging to develop Cafe ICE, a federated learning based imputation method that uses ICE. Cafe ICE is effective for the setting, where data distributed across parties is iid but missing mechanisms are non-MCAR and heterogeneous such that some parties observable data distributions are complementary to the missing data distributions of other clients (as discussed in Section III-C). Algorithm 1 outlines the key steps to implement Cafe ICE, and Algorithm 2 outlines how to compute the Cafe average. Fig. 3 pictorially depicts the key steps taken by *Cafe*-averaging at each iteration.

Cafe ICE (Algorithm 1) begins with each client k having its data $\boldsymbol{X}^k$ (possibly with missing values). In the *setup phase* (Line 1-5), first, the clients compute the set of incomplete features, $\mathcal{F}$ (this can be done by each client sending its incomplete features to the aggregator). Second, each client k computes the missing indicators $\boldsymbol{R}^k$ of its data. Recall that $\boldsymbol{R}^k_{fi} = 1$ if the feature f is missing in the i-th sample in $\boldsymbol{X}^k$ and $\boldsymbol{R}^k_{fi} = 0$ otherwise. Note that, for simplifying the presentation, *the first subscript refers to the feature*.

The last step in the setup phase is the *naive federated imputation* (Line 5). Namely, for each incomplete $f \in \mathcal{F}$, they use the federated average to compute the global mean of f. This can be done as follows. 1) Each client k computes the mean, μ^k_f, using $\boldsymbol{X}^k$; then it sends the μ^k_f and N^k_f to the aggregator (where N^k_f is the number of f-observable samples in $\boldsymbol{X}^k$). 2) The aggregator computes $N_f = \sum_{k=1}^{n} N^k_f$ and sends $\mu_f = \sum_{k=1}^{n}(\mu^k_f \cdot N^k_f)/N_f$ to all the clients. 3) For each incomplete feature f in $\boldsymbol{X}^k$, the client k uses μ_f as the imputed value for all the missing values of f. This results in the first imputation of the data, denoted as $\widetilde{\boldsymbol{X}}^k$. We note that clients could instead use the local mean (i.e., μ^k_f) as the imputed value, the potential implications of which are discussed later. This strategy is useful to limit information disclosure in the case of categorical features where the clients can use local mode as the initial imputation.

After the setup, follows the *iterative imputation phase*, wherein the clients iteratively improve their imputation models. Below, we describe the algorithmic details of this phase for an arbitrary round t and an incomplete feature $f \in \mathcal{F}$.

Clients first need to train the imputation models and mechanism models using their local data (Line 8–12). Specifically, each client k takes the most recently imputed data (either in line 5 or lines 16–17), $\widetilde{\boldsymbol{X}}^k$, and partitions it into $\boldsymbol{A}^k$ and $\boldsymbol{B}^k$ which respectively correspond to the f-observable samples and f-*un*observable samples from $\boldsymbol{X}^k$. The client k builds a local imputation model, $\boldsymbol{\theta}^k_f$, for f by linear regression on the data where the dependent variable $Y = \boldsymbol{A}^k_f$ and the independent variable(s) $X = \boldsymbol{A}^k_{-f}$ (where $\boldsymbol{A}^k_f$ gives the data for feature f, while $\boldsymbol{A}^k_{-f}$ give the data for all the features but f). Next, it builds the model, $\boldsymbol{\xi}^k_f$, for its missing mechanism using logistic regression for the dependent variable $Y = \boldsymbol{R}^k_f$ based on the independent variables $X = \widetilde{\boldsymbol{X}}^k$. It then sends $\boldsymbol{\theta}^k_f$ and $\boldsymbol{\xi}^k_f$ to the aggregator. We argue that our method of approximating the missing mechanisms using logistic regression is very appropriate in the context of ICE imputation, and our empirical evaluation shows that missing mechanism complementarity can indeed be captured by our model (Section I) and serves as a good proxy for mechanism heterogeneity.

After the aggregator receives the models from all the clients, it uses Algorithm 2 to compute the updated models ($\widetilde{\boldsymbol{\theta}^k_f}$ for client k) using Cafe averaging (we will discuss it shortly), and sends the updated models to the clients (Line 13–14).

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Algorithm 1: Cafe ICE.

Input: $\boldsymbol{X}^1, \boldsymbol{X}^2, \ldots, \boldsymbol{X}^n$ (data w/ misses, $\boldsymbol{X}^k$ at client k)
Output: $\widetilde{\boldsymbol{X}}^1, \widetilde{\boldsymbol{X}}^2, \ldots, \widetilde{\boldsymbol{X}}^n$ (imputed data)

// Initial Setup and Imputation
1 $\mathcal{F} \leftarrow$ `Incomplete-Features`$(\boldsymbol{X}^1, \ldots, \boldsymbol{X}^k)$
2 **for** *each client* $k = 1, \ldots, n$, **do**
3 $\quad \boldsymbol{R}^k \leftarrow$ `Missing-Indicator`$(\boldsymbol{X}^k)$
4 **end**
5 $(\widetilde{\boldsymbol{X}}^1, \ldots, \widetilde{\boldsymbol{X}}^n) \leftarrow$ `Naive-Fed-Impute`$(\boldsymbol{X}^1, \ldots, \boldsymbol{X}^n)$
// Iterative Imputation
6 **for** *each round* $t = 1, \ldots, T$ **do**
7 $\quad$ **for** *each feature* $f \in \mathcal{F}$ **do**
$\quad\quad$ // Fit Local Models
8 $\quad\quad$ **for** *each client* $k = 1, \ldots, n$ **do**
9 $\quad\quad\quad \boldsymbol{A}^k, \boldsymbol{B}^k \leftarrow$`Partition`$(\widetilde{\boldsymbol{X}}^k, \boldsymbol{R}^k_f)$
10 $\quad\quad\quad \boldsymbol{\theta}^k_f \leftarrow$`Linear-Reg.fit`$(\boldsymbol{A}^k_f, \boldsymbol{A}^k_{-f})$
11 $\quad\quad\quad \boldsymbol{\xi}^k_f \leftarrow$`Logit-Reg.fit`$(\boldsymbol{R}^k_f, \widetilde{\boldsymbol{X}}^k)$
12 $\quad\quad$ **end**
$\quad\quad$ // Server Aggregation
13 $\quad\quad$ Set $\Theta_f = \{\boldsymbol{\theta}^1_f, \ldots, \boldsymbol{\theta}^n_f\}$ and $\Xi_f = \{\boldsymbol{\xi}^1_f, \ldots, \boldsymbol{\xi}^n_f\}$
14 $\quad\quad$ $\{\widetilde{\boldsymbol{\theta}}^1_f, \ldots, \widetilde{\boldsymbol{\theta}}^n_f\} \leftarrow$ `Cafe-Avg`(Θ_f, Ξ_f) ▷ **Algorithm 2**
$\quad\quad$ // Update Models and Imputation
15 $\quad\quad$ **for** *each client* $k = 1, \ldots, n$ **do**
16 $\quad\quad\quad \boldsymbol{B}^k_f =$ `Linear-Reg.predict`$(\widetilde{\boldsymbol{\theta}}^k_f, \boldsymbol{B}^k_{-f})$
17 $\quad\quad\quad \widetilde{\boldsymbol{X}}^k =$ `Concatenate`$(\boldsymbol{A}^k, \boldsymbol{B}^k)$
18 $\quad\quad$ **end**
19 $\quad$ **end**
20 **end**

Algorithm 2: Cafe Averaging.

Input: $\boldsymbol{\theta}^1, \ldots, \boldsymbol{\theta}^n$ (clients' imputation models) for feature f
$\boldsymbol{\xi}^1, \ldots, \boldsymbol{\xi}^n$ (clients' miss mechanism models) for feature f,
hyper-parameters: α, β, γ
Output: $\widetilde{\boldsymbol{\theta}}^1, \ldots, \widetilde{\boldsymbol{\theta}}^n$ (clients' updated imputation models)

1 $M_f = \max\{N^1_f, \ldots, N^n_f\}$
2 **for** *each* $k = 1, \ldots, n$ **do**
3 $\quad$ // Complementarity and Sample Size Scores
4 $\quad$ **for** *each* $\ell \neq k$ **do**
5 $\quad\quad c^{k\ell} =$ `Complementarity-Score`$(\boldsymbol{\xi}^k, \boldsymbol{\xi}^\ell)$ ▷ **Eq. 1**
6 $\quad\quad s^\ell = N^\ell_f / M_f$
7 $\quad\quad a^{k\ell} = \alpha \cdot c^{k\ell} + (1-\alpha) \cdot s^\ell$
8 $\quad$ **end**
9 $\quad$ // Normalization
10 $\quad$ **for** *each* $\ell \neq k$ **do**
11 $\quad\quad w^{k\ell} = (a^{k\ell})^\beta / \sum_{j \neq k} (a^{kj})^\beta$
12 $\quad$ **end**
13 $\quad$ // Final Aggregation
14 $\quad \widetilde{\boldsymbol{\theta}}^k = \gamma \cdot \boldsymbol{\theta}^k + (1-\gamma) \cdot \sum_{\ell \neq k} w^{k\ell} \cdot \boldsymbol{\theta}^\ell$
15 **end**

Clients then use updated models to do imputation (Line 15-18). Specifically, once each client k receives its updated model $\widetilde{\boldsymbol{\theta}}^k_f$, the updated model is used to compute new imputation values for its missing values of f using the data $\boldsymbol{B}^k$. The new imputed values are stored in $\boldsymbol{B}^k_f$. Next, the client concatenates $\boldsymbol{A}^k$ and $\boldsymbol{B}^k$ (now with updated imputed values) and uses it as $\widetilde{\boldsymbol{X}}^k$, which will be used in going forward (i.e., for the next feature or the next round), or as the final imputation if the iteration procedure is set to terminate (either due to convergence or due to carrying out the sufficient number of rounds.)

Let us now discuss Algorithm 2, which implements the Cafe Averaging algorithm as described in Section III-C. Recall from Section III-C, that for each client k, Cafe averaging weighs the models of the other clients $\ell \neq k$, proportional to their f-observable sample sizes (i.e., N^k_f) and the complementarity of their missing mechanism's models with respect to the missing mechanism model for k. Thus, for each client k, we compute the *complementarity scores* ($c^{k\ell}$ in Line 5) and sample-size scores (s^ℓ in Line 6) corresponding to all the other clients. For complementarity scores (given by (1)), we use the *cosine similarity* between the missing mechanism models, $\boldsymbol{\xi}^k_f$ and $\boldsymbol{\xi}^\ell_f$ (both of which are vectors in $\mathbb{R}^{d+1}$).

$$\texttt{Complementarity-Score} = \frac{1}{2}\left(1 - \frac{< \boldsymbol{\xi}^k_f, \boldsymbol{\xi}^\ell_f >}{||\boldsymbol{\xi}^k_f|| \cdot ||\boldsymbol{\xi}^\ell_f||}\right) \quad (1)$$

where $< \boldsymbol{\xi}^k_f, \boldsymbol{\xi}^\ell_f >$ gives the dot product and $||\boldsymbol{\xi}^k_f||$ gives the euclidean norm of the vector.

We note that the complementarity scores are commutative, and they range between 0 and 1, where 0 means the models are identical, namely, they have no complementarity; and 1 means there is significant complementarity between the models. In the case, when one or both of the clients have no value missing in their feature f, the complementarity score is defined to be 1. This is because all the clients have iid. samples (in our setting), and each one has significant information to contribute when f is fully observable in their data.

After computing the complementarity score of the client ℓ for the client k, we compute the sample size-based score of the client ℓ, which is based on the number of f-observable samples in $\boldsymbol{X}^\ell$ and is computed as 'relative' size, i.e., $N^\ell_f / \max\{N^1_f, \ldots, N^n_f\}$. Here, we use *max* as opposed to *sum* because we want the complementarity and sample size scores to be comparable when we take their average to compute the *combined-score* ($a^{k\ell}$ in Line 7); this lets us control their influence only using the parameter α (as discussed in Section III-C).

Once for a client k, all the combined-scores for all the clients, $\ell \neq k$, are computed, they are normalized (as shown in Line 10, here, we use β to control the normalized distribution). Finally, as discussed in Section III-C, the final updated model for client k ($\tilde{\theta}^k$) is obtained by a weighted combination of its own model (θ^k) and aggregated model from all other clients ($\sum_{\ell \neq k} w^{k\ell} \cdot \boldsymbol{\theta}^\ell$) controlled by the parameter γ as shown in Line 14. This completes the discussion and description of our approach.

E. Hyperparameters

We have three independent hyperparameters in our algorithm α, β, γ. (1) α controls the relative importance / influence of complementarity versus sample-size. (2) γ controls the influence of one's own imputation model w.r.t other clients' imputation model in the updated model. If the clients prefer, they can choose their own γ's (i.e., γ_k for client k); this gives a way to control how much each client wants to weigh its own model towards its updated model. In this work, however, we limit our study to

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TABLE II
DATASET SUMMARY

Datasets	#Samples	#Features	#Classes	#Pos/#Neg
Codon	13,008	26	10	-
Codrna	20,000	9	2	0.53
MIMICIII	20,000	51	2	0.37
Heart	20,000	25	2	0.09
Genetic	20,000	33	2	0.33
HHNP*	87,956	32	2	0.42
eICU*	12,889	24	2	0.09

#Pos/#Neg represents the positive and negative sample ratio, indicating the class imbalance of datasets. * denotes a real-world federated dataset, i.e., with naturally partitioned data.

having the same fixed γ for all. (3) β controls the normalization of weights aggregation. A larger β amplifies the contribution of larger weight components.

By adjusting specific hyperparameters, our algorithm can adapt to various data missingness scenarios. For example, by setting γ, a client can decide how much it relies on other clients. If the client thinks the local missing mechanism is not complicated (say, MCAR), it can set γ to be high to rely more on its local model. α controls the contribution between sample size and complementary score. When α increases to 1, the aggregation weights are only determined by sample size, so it returns to a simple average-based algorithm. Searching for the best strategies for hyper-parameters can be done either by using domain knowledge or some reward-based algorithm (e.g., Reinforcement Learning), which can be an interesting direction for future research.

IV. Empirical Evaluation and Results

In this section, we rigorously evaluate *Cafe* ICE across diverse configurations. We consider practical data federation scenarios, with clients owning iid datasets that have missing values (simulated using heterogeneous MNAR mechanisms), and consider a range of data distribution settings (sample sizes and mechanism choices). We perform imputation in each setting using *Cafe* ICE and other baselines. We evaluate the quality of imputation using a range of metrics as well as downstream federated prediction. Finally, we give an analysis of the evaluation results.

A. Experimental Settings

Datasets: We use seven large standard public datasets from the fields of healthcare and bioinformatics (given in Table II). *Codon:* for codon usage frequencies classification from uci machine learning repository [43]. *Codrna:* for non-coding RNA detection [44]. *MIMICIII:* a dataset of EHRs (electronic health records) from ICUs [45], adapted for patient post-admission modality problem [46], [47]. *Heart:* for heart disease prediction based on indicators extracted from annual CDC surveys [48]. *Genetic:* a dataset curated from ClinVar dataset [49] for genetic variant conflicts prediction. The last two datasets (*HHNP* and *eICU*) are real-world federated datasets. Details about these two datasets (e.g., data pre-processing and client selection) can be found in Section IV-D.

Categorical features were replaced by their one-hot encoding, followed by Principal Component Analysis (PCA) [50], thus resulting in only numeric features. We capped the dataset sizes to be at twenty thousand samples; if a dataset was larger, we obtained a uniform sample.

Baselines: We compare *Cafe* ICE with three ICE-based imputation baselines, as well as two other state-of-the-art imputation models: (1) *Local* imputation, where each client imputes its data using ICE locally (without any collaboration). (2) *FedICE*, federated ICE, which uses model-weights proportional to clients' sample sizes in the federated averaging [8], [33]. (3) *FedMIWAE*, federated averaging version of MIWAE imputation [11], [51]. (4) *FedGAIN*, federated averaging version of GAIN imputation. [52]

Evaluation Metrics: We measure imputation quality using RMSE (root mean square error), Sliced-WS (sliced Wasserstein distance) [53], and visualization of t-SNE (t-distributed Stochastic Neighbor Embedding) plots [54]. Lower RMSE and Sliced-WS denote better imputation quality. We further measure the effects of imputation quality via downstream federated prediction using the imputed datasets, and report the performance using AUROC (area under the receiver operating characteristics) and F1-Scores (we used these metrics because of the class imbalance in the datasets, see Table II, though we also checked standard Accuracy and area under the precision-recall curve, AUPRC, which are not reported due to space limitations.) For the downstream task, a global model (2-layer neural network) was learned using federated learning with FedAvg protocol [27]. The performance is reported over iid sampled test sets.

Implementation Details: We implement *Cafe* ICE[1] by modifying the Iterative Imputer of the Sklearn library [32]. We use ridge and logistic regression for imputation and missing mechanism prediction models with 20 rounds. For *Cafe* ICE, we grid searched the optimal parameter sets (α, β, γ) in parallel with α from 0.5 to 1.0; β from 2 to 4; and γ from 0.05 to 0.3. While using grid search can cost `5-10x` time than that of a single run depending on different datasets, this time can be reduced using advanced tuning methods, such as Bayesian optimization [55], and more importantly, during the experiment, we found that the parameters set (i.e., $\alpha = 0.95$, $\gamma = 0.05$, $\beta = 4$) is applicable across all datasets and scenarios. We use PyTorch 11.2 [56] for federated prediction via FedAvg [27] algorithm. Each client used a two-layer neural network with 32 to 64 hidden nodes (depending on dataset dimension) with the Adam optimizer and learning rate 0.001. Each experiment setup is repeated five times with new and independent randomness to ensure reliability, and the average results are reported.

B. Missing Data Simulation & Federated Scenarios

We consider MNAR mechanisms as they are the most challenging to address in real-world settings [57]. *We choose to use simulations since they allow for a more comprehensive*

[1]Source Code: https://github.com/sitaomin1994/federated_imputation

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evaluation than using real-data with existing missingness (which may be only due to a single mechanism, with no ground-truth to check.) Indeed, simulation allows us to have the ground-truth available for a wide range of missingness, which is created from multiple mechanisms or a single mechanism that due to its inherent randomness produces different missing values at each run. We use two widely used methods to simulate MNAR missing data to construct a variety of heterogeneous settings.

The first one is a quantile-based method [33], [58] to design missing mechanisms as follows. We consider two classes of MNAR missing mechanisms, MNAR-Left and MNAR-Right. For a given feature, f, and a missing rate ρ, an *MNAR-Left* mechanism erases f's values (i.e., replaces by NA) that are in percentile range $[0, \rho]$. On the other hand, *MNAR-Right* erases f's values that are in percentile range $[1-\rho, 1]$. This design allows us to consider a range of mechanisms as well as complementarity relationships.

Using the above classes of missing mechanisms and missing rates, we consider the four different levels of missing mechanism complementarity, and will use probabilistic mix of them to simulate missingness to create different scenarios (S1-S4, discussed shortly). Let us use $\mathcal{M}_1 = \mathcal{M}(D_1, \rho_1)$ and $\mathcal{M}_2 = \mathcal{M}(D_2, \rho_2)$ to denote our missing mechanisms with missing rate ρ_1 and ρ_2 and directions $D_1, D_2 \in \{\text{R}, \text{L}\}$ (R for Right and L for Left).

Perfect Complementarity: We say $\mathcal{M}_1$ and $\mathcal{M}_2$ are in perfect complementarity if $D_1 \neq D_2$ and $\rho_1 = 1 - \rho_2$, e.g., $\mathcal{M}(L, 0.3)$ and $\mathcal{M}(R, 0.7)$ are in perfect complementarity. (Note: here, the part of the distribution one mechanism erases is precisely what is observable under the other.)

Imperfect Complementarity: $\mathcal{M}_1$ and $\mathcal{M}_2$ are in imperfect complementarity if $D_1 \neq D_2$ and $\rho_1 \neq 1 - \rho_2$.

One-sided Complementarity: $\mathcal{M}_1$ and $\mathcal{M}_2$ are in one-sided complementarity if $D_1 = D_2$ and $\rho_1 \neq \rho_2$, e.g., $\mathcal{M}(L, 0.3)$ and $\mathcal{M}(L, 0.5)$.

No Complementarity: $\mathcal{M}_1$ and $\mathcal{M}_2$ have no complementarity if $D_1 = D_2$ and $\rho_1 = \rho_2$, i.e., they are identical. Furthermore, we consider a complex MNAR simulation method using logistic regression [33], [59] – referred to as *Logit based mechanism*. Here, for each feature, its missing values are generated by a logistic model with random weights, which takes the feature itself and the other 25% of highly correlated features of the dataset as inputs. To allow for missing mechanisms heterogeneity and complementarity, for each feature, the random weights for the logistic model are drawn from the region of a unit hypersphere confined to a conical section defined by a specific angular range (30 degrees) along the feature's axis. This constraint maintains the MNAR nature of the missing data mechanism while ensuring that missingness in one feature correlates with other features. Since we use this mechanism to create missing values in all the features, 1) the missingness is MNAR, 2) missingness of the features is correlated with multiple variables, and 3) the missingness is much more complex relative to the approach in [59] which creates only one feature with MNAR missingness.

Federated Learning Scenarios Under Missing Data: We consider ten clients, each with an iid sample as its local data. To simulate the heterogeneity of missing mechanisms among clients, we consider the following scenarios, differing in the levels of complementarity and generalization. For generality of the results, each scenario is created five times, leading to new mechanism and missing values, and the average results are reported. Let us use the $\mathcal{M}$-set to denote the set of all missing mechanisms corresponding to different values of ρ's and direction.

Ideal scenario: For all features, half of the clients are $\mathcal{M}(L, 0.5)$ and the other half are $\mathcal{M}(R, 0.5)$, creating an ideal scenario of complementarity, where *Cafe* should perform very well. *Perfect complementarity scenarios* **(S1)**: Pick a client uniformly at random, then for each of its features, pick a missing mechanism uniformly at random from $\mathcal{M}$-set and assign the perfect complement of the mechanism for the same feature for all other clients. For example, if for a feature, f, $\mathcal{M}(L, 0.3)$ was picked for the selected client, then all others will have $\mathcal{M}(R, 0.7)$ for f. This scheme produces a complex mix of missing mechanisms to create missing values in the data such that each feature has missingness due to different mechanisms, and different features have different partitions of clients as complementary. *Imperfect complementarity scenario* **(S2)**: Similar to S1, pick a client at random, then for each of its features, pick a mechanism at random from $\mathcal{M}$-set for it, and for each of the other clients, pick at random a mechanism with imperfect complementarity. *One-sided complementarity scenario* **(S3)**: It is similar to S2 except for the other clients, we pick at random a mechanism with one-sided complementarity. *No complementarity scenario* **(S4)**: All clients have the same mechanism for all the features, i.e., either $\mathcal{M}(R, 0.5)$ or $\mathcal{M}(L, 0.5)$, resulting in absolutely no complementarity.

Complex mixture scenarios: Additionally, we consider two complex scenarios consisting of a mixture of missing mechanisms, varying across features and clients with respect to complementarity and missingness correlation with other features. In *Complex Scenario# 1*, for each client and each feature, the quantile-based mechanism is randomly picked from the $\mathcal{M}$-set, resulting in the missing data being independent of the other features. *Complex Scenario# 2* follows the first scenario except the $\mathcal{M}$-set now consists of the Logit-based mechanism (described above), which is used to simulate missing values (correlated with multiple features) for different missing ratios. Now, consider the missing rate, ρ: although ρ can take any value between 0 and 1, we limit the missing rate to be moderate, i.e., in between 0.3 and 0.7 because we want to have sufficient data (for learning) and sufficient missing rate (to analyze its effects) in all of our cases. Under these limits, we consider $\rho = 0.3, 0.4, 0.5, 0.6, 0.7$, i.e., increments of 0.1. Now, $\mathcal{M}$-set will consist of mechanisms corresponding to these missing rates for each direction. We also evaluated using more granular ρ's and found the results to be similar.

C. Results

Our evaluation of *Cafe* focuses on the following aspects and results in the following key observations:

1) *Complementarity score verification: Cafe* can effectively capture complementarity between missing mechanisms across clients under different scenarios.

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Fig. 4. Pairwise Complementarity. The heatmap plots the pairwise complementarity scores for missing mechanism models across clients (for Codon, exemplified by one feature; other features have similar results). From Ideal to S4, the level of complementarity decreases. Here, blue hues indicate high complementarity, red indicates pretty low complementarity, and yellow and green hues indicate moderate levels.

2) *Performance across complementarity levels:* Under high complementarity levels, Cafe significantly outperforms the other baselines. Otherwise, it performs as well as others.
3) *Effectiveness in complex scenarios: Cafe* performs well in complex settings.
4) *Impact of sample size distribution: Cafe* is robust, i.e., its performance remains consistent under different client sample-size distributions.
5) *Scalability and efficiency:Cafe*'s performance is comparable to that of the baselines for different numbers of clients and sample sizes.

Complementarity scores evaluation: We first show that our method to compute complementarity scores, which we use in *Cafe* ICE, can effectively capture the different levels of complementarity between clients' missing mechanisms. As illustrated in Fig. 4, the heatmap of complementarity scores reveals that the distinct patterns across different scenarios are captured correctly. In the Ideal and Perfect complementarity scenarios (S1), a clear distinction in complementarity scores is observed between clients, with extremely high scores (close to 1.0) indicating perfect complementarity and low scores (close to 0) within clients of the same mechanism. Conversely, No complementarity scenario (S4) shows uniformly low scores, reflecting the absence of complementarity. In the Imperfect (S2) and One-sided complementarity scenarios (S3), complementarity scores show variation, with some clients exhibiting relatively high but not near-perfect scores, indicative of some degree of complementarity. Overall, our scoring scheme accurately reflects the complementarity level of clients' missing mechanisms in each scenario.

Evaluation across different levels of missing mechanism complementarity: In this section, we conduct a comprehensive evaluation of *Cafe* ICE under scenarios with various levels of client missing mechanism complementarity. Fig. 5 illustrates imputation quality and federated prediction results. We make three key observations. First, there is a trend of decreasing imputation performance by *Cafe* ICE as the level of missing mechanism complementarity reduces, aligning with our earlier findings on complementarity score verification. Second, in scenarios with high to moderate complementarity (i.e., *Ideal*, **S1**, and **S2**), *Cafe* ICE significantly outperforms baselines in both imputation and federated prediction tasks. Third, in scenarios with lower complementarity (i.e., **S3** and **S4**), *Cafe* ICE maintains comparable imputation quality and federated prediction performance to the baselines. Overall, these observations support *Cafe* ICE as a more robust and adaptable solution compared to the state-of-the-art baselines.

Evaluation in complex scenarios: In practice, we expect the missing scenarios are complex. Here, we consider complex scenarios (#1 and #2 as discussed above) where missing mechanisms are more sophisticated (e.g., logit-based mechanism) and are a mix of varying levels of complementarity across features and clients (here, all clients have the same number of samples). The evaluation focuses on imputation quality (RMSE), convergence rate, and downstream task performance.

1) Imputation quality and convergence: The RMSE of imputation for *Cafe* ICE and the baselines are shown in Fig. 6. Across both complex scenarios, *Cafe* ICE achieves significantly lower errors compared to not only ICE-based baselines but also VAE or GAN-based advanced imputation methods. t-SNE plots (Fig. 7) also show the same; that is, the data imputed by *Cafe* ICE closely aligns with the ground truth, unlike other methods, which exhibit noticeable deviations, underscoring *Cafe*'s accurate recovery of missing values. Next, we examine the error convergence (Fig. 8), observing all methods converge within 5-10 iterations. Notably, *Cafe* ICE's error diminishes significantly upon convergence, in contrast, the baselines can even result in higher error. Upon convergence, the final imputation error of *Cafe* ICE is significantly lower than baselines. We also note that individual clients' performance mirrors the general trend as seen from the narrow error bands around the solid line depicted in Fig. 8.

2) Federated prediction: Fig. 9 shows that overall *Cafe* ICE outperforms the baselines, particularly for Codon, Codrna, and MIMICIII. *Cafe* ICE performs significantly better on the Codon task, a multi-class classification task, which is more sensitive to missing data than other tasks. Even if the baselines (*FedICE* or computationally demanding *FedGAIN* and *FedMIWAE*) can approach *Cafe* ICE in some cases, they exhibit higher variation across multiple runs of the experimental (depicted by the black). Baselines' higher variation indicates their sensitivity to small distributional differences due to randomness (e.g., a different seed). Thus, these results indicate *Cafe* ICE's robustness and effectiveness in enhancing downstream federated prediction in practical settings.

Effect of varying sample sizes across clients: Here, we consider complex scenarios for missing data simulation to avoid bias towards any particular method. First, we explore the effect of sample sizes across clients. We use three strategies to select sample sizes for each client and then assign to each client the iidsamples corresponding to its selected size. The following are the three strategies. *Uni* assigns equal sample sizes; *Dir* assigns sample sizes according to Dirichlet distribution (with parameter 0.1 for moderate heterogeneity) [60]; and *HS* (hub-spoke setting) assigns one client (hub) 50% of total data and the rest (peripherals) 5% each. Our results, focusing on Sliced-WS and AUROC (with RMSE and F1 showing similar trends), are given in Fig. 10. Although *FedICE* takes the sample size into account, it is unable to capture the effect of heterogeneity or deal with the impact of sample size on it. Conversely, *Cafe* ICE consistently exhibits minimal variation and surpasses the baselines. Notably, *Cafe* ICE (compared to the baselines) shows

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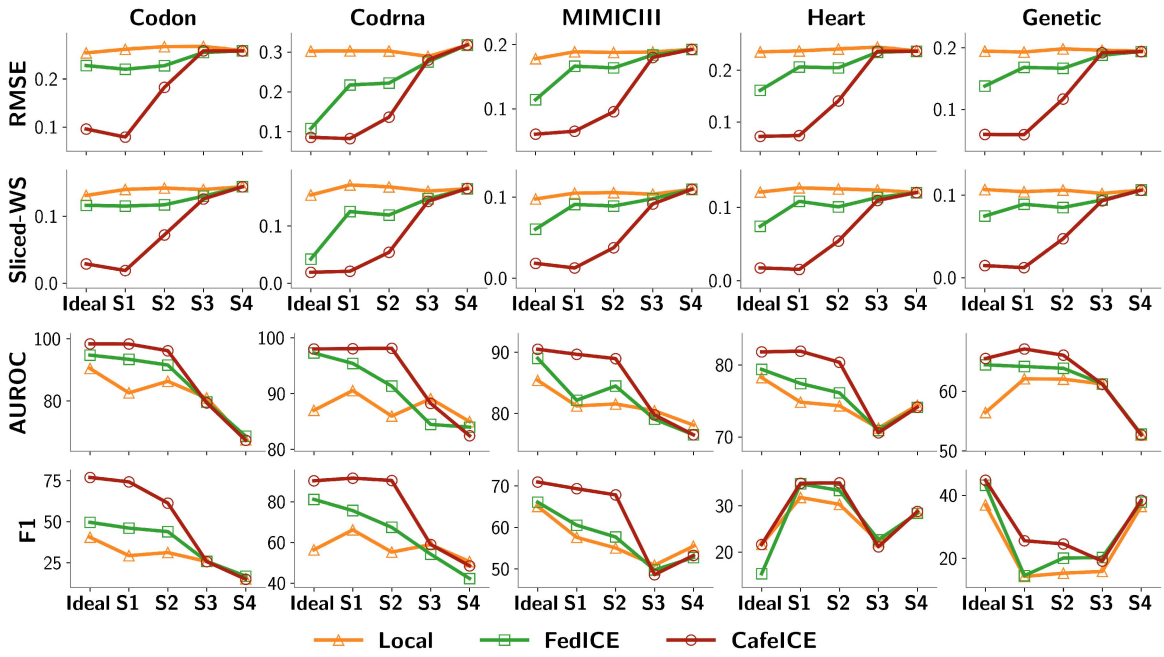


Fig. 5. Evaluation on Different Levels of Missing Mechanism Complementarity. Scenarios (Ideal, S1 - S4) are ordered in decreasing levels of missing mechanism complementarity.

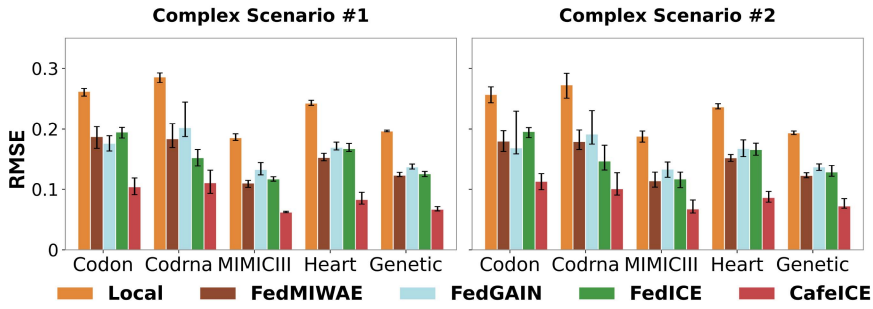
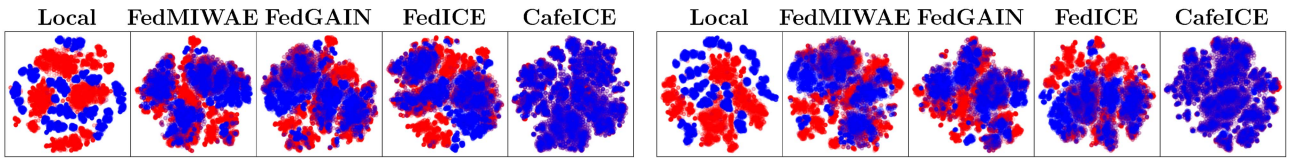


Fig. 6. RMSE. Federated Imputation Performance (complex scenarios).



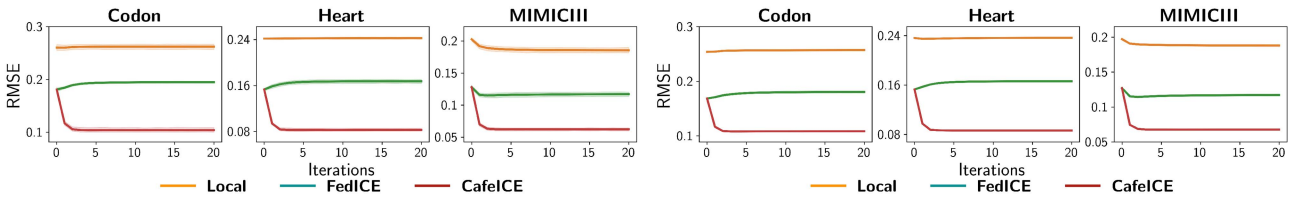
(a) **Complex Scenario# 1 (quantile based mechanisms)** (b) **Complex Scenario# 2 (logit based mechanisms)**

Fig. 7. t-SNE Visualization for Imputed versus Ground-truth Data. Each subplot presents a scatter plot projection of the ground-truth data (red points) and the imputed data (blue points) over the merged data from all clients for the Codon task. A greater overlap between red and blue points signifies better imputation quality.

improved performance when sample sizes differ across clients, indicating its effectiveness in realistic scenarios.

*Effect of varying the number of clients:*We now investigate the impact of varying numbers of perfect complementarity clients on *Cafe* ICE's imputation performance. We simulate a scenario with one hub client as MNAR-Left ($\mathcal{M}(L, 0.5)$) and an increasing number of clients exhibiting perfectly complementarity mechanisms (i.e., MNAR-Right as ($\mathcal{M}(R, 0.5)$) from 2 to 10. As depicted in Fig. 11, the presence of complementarity clients significantly reduces imputation errors, and this effect remains stable with an increasing number of complementarity clients. This observation suggests that *Cafe* ICE's performance is

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(a) **Complex Scenario# 1 (quantile based mechanism)** (b) **Complex Scenario# 2 (logit based mechanism)**

Fig. 8. Imputation Error Convergence Curve. Each line depicts the average imputation error across all clients.

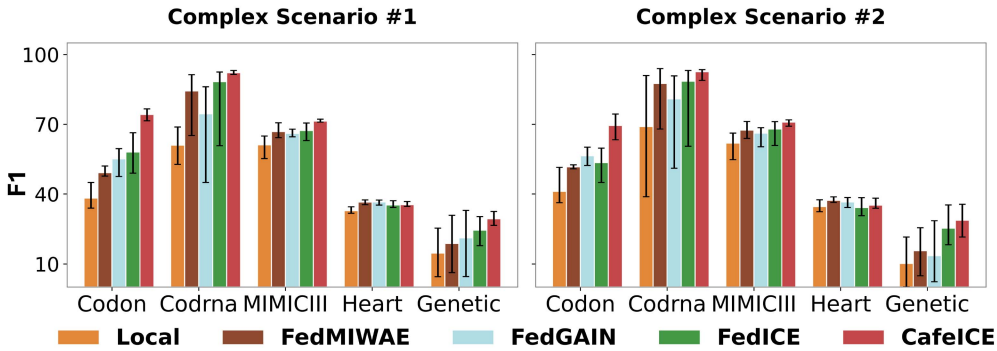


Fig. 9. F1 Score. Federated Prediction Performance (complex scenarios).

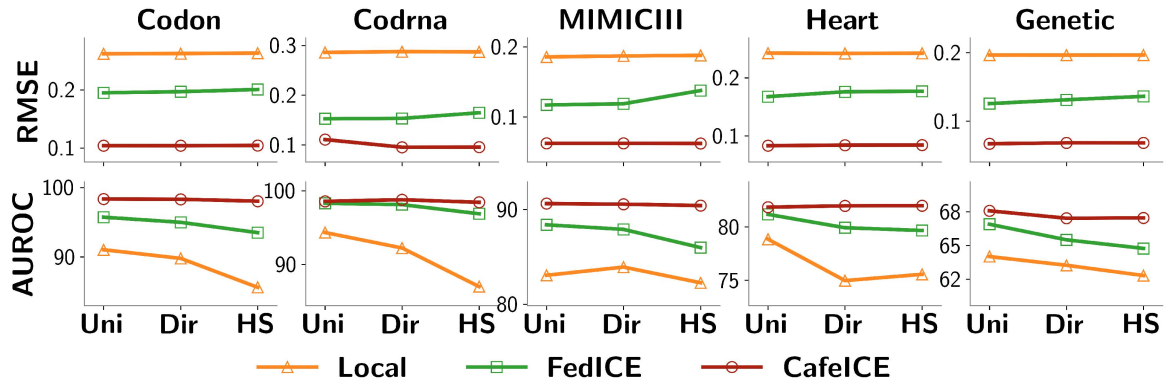


Fig. 10. Evaluation on Different Sample Size Strategies.

TABLE III
RUNNING TIME (S)

Methods	Codrna	Codon	MIMICIII	Heart	Genetic
Local	120.79	160.88	654.9	271.64	330.55
FedICE	126.09	160.15	688.75	281.33	341.17
***Cafe* ICE**	126.48	163.78	744.06	293.19	370.99

Average accumulated running time for all clients and the server across all scenarios.

robust to the number of complementarity clients. *Particularly, a few clients with complementarity missingness can markedly enhance the effectiveness of imputation*. Notably, *Cafe* ICE's performance remains stable as the data and network size grows.

Evaluation on running time: Finally, we compare *Cafe* ICE's runtime with baseline methods. We report the average running time for each dataset across all scenarios for four methods. As shown in Table III, given that *Cafe* ICE only requires fitting an additional linear model for mechanism prediction, its runtime for each dataset is comparable to that of *FedICE* and other baselines. Overall, *Cafe* ICE is quite efficient in practice.

D. Evaluation on Real World Federated Data

In this section, we evaluate *Cafe* ICE on two real-world federated datasets from multiple healthcare providers or hospitals to demonstrate its practical applicability in the real world. We selected participating silos based on pairwise sliced Wasserstein

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TABLE IV
EVALUATION RESULTS ON REAL WORLD FEDERATED DATA

Methods	HHNP						eICU					
	Imputation Quality		Downstream Prediction				Imputation Quality		Downstream Prediction			
	RMSE	Sliced-WS	ACCU	F1	AUROC	AUPRC	RMSE	Sliced-WS	ACCU	F1	AUROC	AUPRC
Local	0.355	0.183	81.5	69.9	87.9	74.8	0.156	0.086	94.3	60.7	89.3	65.1
FedMIWAE	0.282	0.136	78.5	62.2	83.6	66.3	0.118	0.060	94.3	63.2	90.6	68.1
FedGAIN	0.272	0.124	81.3	68.7	87.2	73.7	0.108	0.054	94.3	64.4	90.6	69.4
FedICE	0.183	0.080	83.2	72.1	90.0	78.8	0.094	0.047	94.7	65.6	90.9	70.3
***Cafe* ICE**	**0.129**	**0.041**	**86.3**	**76.0**	**92.5**	**84.5**	**0.065**	**0.020**	**95.2**	**68.5**	**92.1**	**73.2**

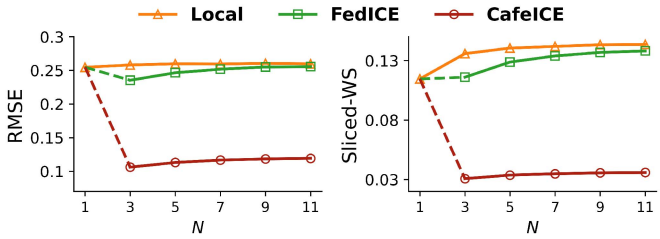


Fig. 11. Evaluation with Varying Numbers of Perfectly Complementary Clients. Each line represents the average imputation errors. Here one client has MNAR-Left ($\rho = 0.5$) and $N - 1$ have MNAR-Right ($\rho = 0.5$). For $N = 1$, the error is for local imputation alone.

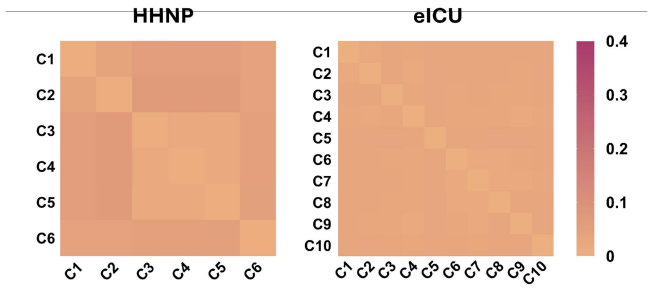


Fig. 12. Client Data Similarity. Sliced Wasserstein distances between clients' local data distributions on HHNP and eICU, with darker colors indicating larger distributional discrepancy.

distances, choosing those with relatively similar data distributions, as shown in Fig. 12. Then, the heterogeneous missing data is introduced in clients following the same setting of Complex Scenario #1 in Section IV-B. The evaluation process is the same as Section IV-A. The details of these two federated datasets are given below.

Heritage Health Provider Networks (HHNP) Dataset: This dataset is provided by Heritage Provider Network, containing patient admission from multiple healthcare providers from California in 2010-2013. We use 32 features, including patient demographics, laboratory results, etc., to predict patients' claim truncation likelihood. We select 6 healthcare providers with similar data distances for evaluation.

Philips eICU Collaborative Database: This database comprises patient admissions from critical care units across 208 U.S. hospitals from 2014 to 2015. We conducted mortality prediction using 24 features, including patient demographics and vital signs based on standardized data preprocessing used in previous studies [61]. Similarly, we selected 10 hospitals that have similar data distributions for evaluation.

The evaluation results are shown in Table IV. The results demonstrate that *Cafe* ICE consistently outperforms baselines on both datasets. On HHNP, it reduces RMSE by 29.5% (from 0.183 to 0.129) compared to second-best method *FedICE*, with similar improvements observed in sliced Wasserstein distance. For downstream prediction, *Cafe* ICE also shows notable gains across all metrics, particularly with F1 score improving by 3.9% (from 72.1% to 76.0%) and AUPRC increasing by 5.7% (from 78.8% to 84.5%) compared to second-best method. Similar improvements are also observed in the eICU dataset. These results validate Cafe ICE's effectiveness in handling heterogeneous missing data across silos in real-world federated healthcare settings.

V. RELATED WORK

Missing Data Imputation: Missing data is a pervasive problem in real-world settings and has attracted significant attention in the data engineering, statistics, and Machine Learning literature [62]. To address the missing data problem, imputation techniques have been developed, including generative, hot-deck-based, and discriminative methods [13], [62], [63].

Generative approaches model the joint feature distributions through methods such as Expectation-Maximization (EM), Markov Chain Monte Carlo (MCMC) [64], and Generative Adversarial Networks (GANs) [52], [65], [66]. Hot-deck imputation [67], utilizing methods like k-Nearest Neighbors (KNN) [68] or Graph Neural Networks [69], substitutes missing entries with those from analogous records. Recent advancements by Large Language Models (LLM) [70] and Optimal Transport [59], [71] have set new benchmarks in imputation accuracy. Compared to the other methods, discriminative techniques, exemplified by the Multiple Imputation by Chained Equations (MICE) approach [34], infer missing values by modeling the conditional distributions of each feature and make no assumptions about data distribution, allowing it to be more adaptable to complex real-world data. Moreover, it can seamlessly integrate with many machine learning models, such as linear regression models and neural networks [34], allowing practitioners to choose the underlying model based on computational demands and the utility required in different practical scenarios. There are many recent research efforts to extend MICE to improve the imputation quality [33], [72], [73]. This study leverages

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the versatility of the MICE framework and proposes a novel adaptation for the federated settings.

Federated Missing Data Imputation: Most of current FL studies primarily focus on using federated learning algorithms for prediction tasks [74], [75], [76]. As for missing data imputation, federated learning offers a solution by facilitating the use of additional datasets, enriching the imputation process with more information. However, existing work typically focuses on adapting conventional methods to federated environments without addressing the complexity of missing data mechanisms. For example, the Expectation-Maximization (EM) method has been expanded to create a federated optimization for biodiversity monitoring imputation tasks [77]. Recent efforts have leveraged deep learning models for federated imputation. Zhou et al. [9] developed a conditional GAN imputation method within a federated learning framework using fedavg [27], addressing the diverse challenges posed by air quality datasets from various sources. Yao and Zhao [10] proposed a GAN-based federated imputation using federated transfer learning, where a local model trained with GAIN [52] on an edge device is refined with knowledge from multiple devices. Balelli et al. [11] adapted the MIWAE imputation model [51] for federated settings using aggregation strategies such as fedavg and fedprox [27], [78]. Typically, these studies presume that the missing data is either MCAR or MAR, and they adopt simple aggregation strategies that fail to address the complexity of missing data patterns across different clients.

The only work considering adapting MICE imputation into the distributed setting is by Chang et al. [8]. However, their approach is limited to simpler missing data assumptions (MCAR or MAR) and does not tackle the intricate MNAR scenarios that may occur among clients. Overall, as we menthoned in Section I, existed works in federated imputation predominantly focus on simple missing mechanisms (MCAR, MAR) with limited focus on MNAR mechanism and, more importantly, they neglect missing data heterogeneity across silos.

VI. Discussions and Conclusion

In this paper, we have proposed a novel complementarity adjusted federated *Cafe* ICE averaging approach for federated imputation that can be used to develop personalized imputation models for different clients. Our approach is highly effective in cases where the underlying missingness is heterogeneous and also performs as well in other scenarios with simpler (MCAR and MAR) missingness characteristics.

With respect to theoretical analysis, extensive analysis has been conducted on conventional FL that facilitates learning a single global model from homogeneous data across clients, under a strongly convex loss function, and several other assumptions, ensuring the convergence of the FedAVG algorithm ([42], Theorem 3.2.) However, in scenarios characterized by complete heterogeneity, local models can diverge significantly, and thus, independently learned local models outperform the global model ([42], Theorem 3.2.) These two extreme cases are special cases of *Cafe* ICE (with specific α, γ values), where the above guarantees would apply. The general case is significantly more complex to analyze since all of the underlying assumptions of convexity, smoothness, and Lipschitz continuity [42], [79] may not apply, contingent on the employed imputation techniques. We leave this to future work.

Although our work focuses on a single imputation based on ICE, it can be easily extended to multiple imputation based on MICE. One intuitive way is to replace the vanilla linear imputation model (such as Ridge Regression) with the Bayesian linear imputation model (such as Bayesian Ridge Regression). The final imputation values can be drawn from the posterior distribution modeled by Bayesian models, which makes multiple imputations possible. The selection of proper prior and optimization strategies for the Bayesian model for each client can also rely on a missing mechanism model, and we plan to study how to design a practical algorithm for this.

We also plan to extend our approach to provide explicit guarantees of privacy in the future.

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Sitao Min is working toward the PhD degree with the Department of Management Science and Information Systems, Rutgers University. His research primarily focuses on data management, data quality, missing data mitigation, and federated learning.



Hafiz Asif received the PhD degree from Rutgers University. He is an assistant professor with Information Systems and Business Analytics Department, Hofstra University, New York, USA. His research is developing SAFE (secure & private, auditable, fair, and equitable) theoretical approaches and practical frameworks to leverage the power of data to solve real-world problems.



Xinyue Wang received the PhD degree from Rutgers University. She is developing privacy-preserving methods to generate high-fidelity synthetic data and analyze genomic data.



Jaideep Vaidya (Fellow, IEEE) is a distinguished professor of computer information systems with Rutgers University. He has published more than 200 papers in international conferences and journals. His research interests are in privacy, security, data management, and data analytics. He is a fellow of the AAAS, ACMI, IAHSI, and IFIP as well as an ACM distinguished scientist.

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